

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-056

WASTE DISCHARGE REQUIREMENTS
FOR
RITE OF PASSAGE ATHLETIC TRAINING CAMPUS
SIERRA RIDGE WASTEWATER TREATMENT PLANT
CALAVERAS COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. Rite of Passage Athletic Training Campus (ROPATCS) submitted a Report of Waste Discharge (RWD), dated 25 October 2000, for the operation and upgrade of the Sierra Ridge Wastewater Treatment Plant (WWTP). Additional information was received on 7 December 2000. ROPATCS is hereafter referred to as Discharger.
2. The WWTP is on Assessor's Parcel Numbers 38-008-010, -11, -12, and -013, and is owned and operated by the Discharger. The WWTP is at 10400 Fricot City Road approximately ten miles southeast of the town of San Andreas, in Sections 27 & 28, T4N, R13E, MDB&M, as shown on Attachment A, which is attached hereto and made part of the Order by reference.
3. ROPATCS is a juvenile rehabilitation facility comprising 250 acres. The facility will be operated by Rite of Passage, a Nevada-based company that provides academic and athletic programs for disadvantaged youths. The 250-acre campus consists of a self-contained community (e.g. administration center, nursing facility, kitchen, dinning hall, dormitories, staff housing, gym, maintenance building, onsite water supply well, and a wastewater treatment facility).
4. Order No. 97-075, adopted by the Board on 25 April 1997, prescribes requirements for the WWTP. This Order is neither adequate nor consistent with the current plans and policies of the Board.
5. WDRs No. 97-075 permitted an average dry weather flow of 12,500 gallons per day (gpd) based on a campus population of approximately 100 people assuming 125 gallons per day per person. The WDRs prescribed treatment of wastewater through an Imhoff pump/surge tank, followed by recirculating intermittent sand filters prior to storage in a reservoir and discharge to a leachfield. The leachfield, although presently still in use, was constructed as a temporary wastewater disposal system.
6. The Discharger has requested to use the leachfield as a temporary emergency disposal option to prevent potential overflows from the storage reservoir. This Order incorporates a flow limit to the leachfield in accordance with the design capacity of the field.
7. The Discharger is in the process of upgrading and expanding the existing WWTP. The upgraded WWTP will consist of a package wastewater treatment unit providing flow

equalization, extended aeration, clarification, disinfection, and sludge storage. The disinfection system will consist of a sodium hypochlorite feed system including duplex feed pumps and a 40-foot long, two-foot diameter chlorine contact chamber. The design capacity of the WWTP is approximately 30,000 gpd.

8. Treated wastewater will be stored prior to disposal via spray irrigation in a newly constructed, 2.2 million-gallon reservoir. The five acre designated sprayfield area is located upgradient and adjacent to the large storage reservoir. The reservoir and sprayfield have been designed with tailwater trenches to return collected tailwater runoff from the sprayfield back into the storage reservoir. During the wet-weather season, the trenches will operate as stormwater diversion trenches to divert surface runoff around the storage reservoir.
9. An existing 0.600 million-gallon storage reservoir located directly below the leachfield is also planned for use after upgrading activities, including lining the reservoir with a 36-mil fully encapsulated fabric pond liner, are complete.
10. The new storage reservoir has been constructed through approximately 18" of soil followed by 25 feet of fractured bedrock. During construction, the bottom of the reservoir intercepted the groundwater table. Staff has informed the Discharger that it must adequately line the entire pond so that groundwater will not be impacted from the storage of wastewater. The pond bottom must first be re-worked to remove a 20' almost shear "bench". The grading and lining activities will reduce the capacity of the pond, so the Discharger must recalculate the capacity of the pond.
11. Increases in the number of students and staff at the ROPATCS facility is planned to occur in stages with corresponding expansions of spray disposal fields. The first stage is based on a campus population of 110 students and staff total. The maximum wastewater flow is expected to be 11,000 gpd assuming a wastewater generation rate of 100 gallons per day per person. For this first stage, a spray disposal field of approximately 5 acres will be developed. Additional spray disposal acreages of 10 acres, 12 acres, and 7 acres have been identified on-site and will be reserved for future development. The sizing of the sprayfields is based upon an applied hydraulic load of 4 feet per year (average 4 inches per month).
12. In accordance with the provisions of the California Environmental Quality Act (CEQA), the Calaveras County Planning Commission approved a Negative Declaration in the adoption of the Conditional Use Permit (CUP 94-55) for the proposed operation of the campus. CUP 94-55 limits the student population to a maximum of 100; therefore this Order contains the same limit assuming 10 staff for 100 students for a maximum population of 110.
13. This Order only allows wastewater generation from a total of 110 students and staff. The Discharger must submit a new RWD, including an appropriate California Environmental Quality Act (CEQA) document, before the Board will consider allowing an increase in the wastewater flows.

14. This permit only authorizes wastewater storage in the 0.6 million gallon pond and the new storage pond. Wastewater may only be discharged to the 5 acre sprayfield upslope of the new storage pond or to the leachfield in a temporary emergency. The Discharger must submit a new report of waste discharge, including the appropriate CEQA document, before the Board will consider allowing additional areas for wastewater storage or disposal.
15. The Discharger proposes to upgrade the wastewater collection system to reduce inflow and infiltration as the campus population expands and additional renovated housing facilities are brought on-line. This Order contains a Provision for annual updates of collection system improvements.
16. The site lies within the South Fork Calaveras hydrologic unit/area/sub-area No. 533.30, as depicted on interagency hydrologic maps prepared by the Department of Water Resources. Surface water drainage is to San Antonio Creek, a tributary to the Calaveras River.
17. The beneficial uses of the Calaveras River are contact and non-contact recreation; warm and cold freshwater habitat; warm water migration, warm and cold water spawning habitat; wildlife habitat.
18. Water for domestic use is supplied by an on-site well that is operated by the Discharger. The well is approximately 2000 feet northeast of the effluent storage reservoir and about 100 feet higher in elevation. The depth of the well is 360 feet and discharges at a rate of 87 gpm. Water-bearing rock is encountered at a depth of 140 feet with a draw-down during pumping to approximately 220 feet.
19. The beneficial uses of underlying groundwater are domestic, industrial and agricultural supply.
20. Based on the Discharger's RWD, groundwater quality can be characterized as follows:

<u>Parameter</u>	<u>Concentration</u> (mg/l)
Carbonate	5
Bicarbonate	109
Total Alkalinity	109
Total Hardness	142
Total Dissolved Solids	258
Nitrate	ND
Chloride	29
Sulfate	45
Calcium	45
Magnesium	7
Sodium	32
Total Iron	0.46
Conductivity (µmho/cm)	322

21. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et. seq., for the use of reclaimed wastewater, and has developed guidelines for specific use. These requirements implement the reclamation criteria in Title 22.
22. Surrounding land uses are primarily residential and rural properties.
23. Federal regulations for stormwater discharges were promulgated by the U.S. Environmental Protection Agency on 16 November 1990 (40 CFR Parts 122, 123, and 124). The regulations require specific categories of facilities which discharge stormwater associated with industrial activities to obtain NPDES permits. The flow at this wastewater treatment plant is less than 1.0 mgd and therefore the Discharger is not required to apply for a stormwater NPDES permit.
24. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basin. These requirements implement the Basin Plan.
25. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
26. Section 13267(b) of California Water Code provides that: “In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports”.
27. The Board has considered anti-degradation pursuant to State Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to determine whether the storage of wastewater will cause an increase in groundwater constituents above that of background levels. If the storage is causing such an increase, then the Discharger may be required to line the ponds, implement source control, change the method of storage, or take other action to prevent groundwater degradation.

28. The Board has notified the Discharger, and interested agencies and persons, of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
29. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 97-075 is rescinded and the Right of Passage Athletic Training Campus, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

Note: Other prohibitions, conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Surfacing of wastewater outside or downgradient of any surface impoundment, the leachfield, or tailwater collection pond is prohibited.
7. As of **15 May 2001**, discharge to the leachfield is prohibited except in temporary emergency situations when overflow from the storage reservoirs is imminent.
8. Application of wastewater to areas different than those described in Finding No. 8 is prohibited.
9. Discharge of untreated or partially treated waste to groundwater is prohibited.
10. The discharge of any wastewater other than that from domestic sources is prohibited.

11. As of **15 March 2001**, additional wastewater shall not be discharged to the 0.600 million-gallon storage reservoir until the Executive Officer has approved the upgrade report required by Provision G.1.g. Wastewater stored in this reservoir prior to 15 March 2001 must be removed by **1 June 2001**.
12. The new storage reservoir may not be used to store wastewater until the Executive Officer has approved the lining report required by Provision G.1.b.
13. The designated sprayfield shall not be used to dispose of wastewater until the Executive Officer has approved the construction report required by Provision G.1.d.

B. Discharge Specifications:

1. The 30-day average dry weather discharge shall not exceed 11,000 gpd.
2. Discharge to the leachfield in temporary emergency situations shall not exceed 2,700 gpd.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the WWTP.
4. As a means of discerning compliance with Discharge Specification No.3, the dissolved oxygen content in the upper zone (1 foot) of the wastewater storage reservoirs shall not be less than 1.0 mg/l.
5. The wastewater storage reservoirs shall not have a pH of less than 6.5 or greater than 8.5.
6. The Discharger shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.
7. The storage reservoirs shall be managed to prevent the breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the waste surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and/or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
8. The freeboard in each storage reservoir shall never be less than two feet as measured vertically from the water surface to the lowest point of overflow.
9. The wastewater treatment and storage system shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual

precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.

10. On or about **1 October** each year, the available storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. 8 and No. 9.
- 11 A licensed wastewater treatment plant operator shall oversee operation and maintenance of the wastewater treatment plant.

C. Effluent Limitations:

The discharge of treated wastewater to the storage reservoir in excess of the following limits is prohibited:

<u>Constituent</u>	<u>Unit</u>	<u>Monthly Average</u>	<u>Monthly Median</u>	<u>Monthly Maximum</u>
Total Coliform Organisms	MPN/100ml	-	23	240
Settleable Solids	ml/l	0.2	-	0.5
Total Dissolved Solids	mg/l	550	-	-
BOD ¹	mg/l	40	-	80

¹ 5 Day, 20° Celsius Biochemical Oxygen Demand

D. Reclamation Requirements

The following requirements apply to the disposal of reclaimed wastewater on ROPATCS property:

1. Reclaimed wastewater shall meet the criteria contained in Title 22, Division 4, CCR (Section 60301, et.seq.) and shall be discharged in accordance with the Wastewater Disposal Operations Plan required by Provision No. G.1.d.
2. The following setback/buffer zones shall be provided at the reclamation areas:

<u>Setback Definition</u>	<u>Irrigation Setback (ft)</u>
Disposal field to public roads	50
Disposal field to property lines	50
Disposal field to domestic well	500
Disposal field to any on-site irrigation well	100
Disposal field to any food crop	100
Disposal field to any surface water drainage course	100

3. Public contact with reclaimed wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

4. To alert the public of the use of reclaimed wastewater, signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal.
5. For areas that may be accessible to the public, the following must be implemented: Reclaimed wastewater conveyance lines shall be clearly marked as such. Reclaimed wastewater controllers, valves, etc., shall be affixed with reclaimed wastewater warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in such a manner, that permits operation by authorized personnel only.
6. All reclaimed wastewater equipment, pumps, piping, valves, and outlets shall be appropriately marked to differentiate them from potable facilities. The reclaimed wastewater piping system shall not include any hose bibs.
7. Areas irrigated with reclaimed wastewater shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within 48-hours.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and un-pressurized pipelines and ditches that are accessible to mosquitoes shall not be used to store reclaimed wastewater.
8. Reclaimed wastewater used for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the sprayfield.
9. Direct or windblown spray shall be confined to the designated disposal area and prevented from contacting drinking water facilities.
10. The Discharger shall not irrigate with effluent 24 hours before precipitation, during periods of precipitation, for 24 hours after cessation of precipitation, or when the ground is saturated.
11. The Discharger may not spray irrigate with effluent when wind velocities exceed 30 mph.
12. The tailwater recapture system must be operated to capture all reclaimed water runoff, as well as any storm water runoff that occurs within 24 hours of the last application of reclaimed water.
13. There shall be no impoundment of reclaimed wastewater within 500 feet of any domestic water well unless it is demonstrated to the satisfaction of the Executive Officer that a shorter distance is justified.

E. Solids Disposal Requirements:

1. Collected screenings, grit, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Storage, use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503 and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).
3. Sludge and other solids shall be removed from storage reservoirs, clarifiers, settling basins, etc., as needed to ensure optimal plant operation and adequate hydraulic capacity. Drying operations shall take place such that leachate does not impact groundwater or surface water.
4. If biosolids will be stored onsite between 15 October and 15 May of any year, then they shall be stored in a facility constructed in accordance with Class II surface impoundment or waste pile standards contained in Title 27 of the CCR, or similar facility approved by the Executive Officer. Such a facility shall be designed and maintained to prevent inundation or washout from a storm or flood with a 100-year return frequency. The Discharger shall collect any leachate or stormwater that comes in contact with the biosolids pile and return it to the wastewater treatment plant.
5. Biosolids may not be disposed of onsite except as a soil amendment used at agronomic rates. Before onsite disposal is permitted, the Discharger must submit a technical report containing all the information required for a Notice of Intent for Water Quality Order No. 2000-10-DWQ (or any subsequent document which replaces Order No. 2000-10-DWQ). The technical report must be submitted at least 60 days before the anticipated date of disposal.
6. Disposal of biosolids at a permitted municipal solid waste landfill or at a permitted publicly owned treatment works is acceptable. The Discharger may also elect to dispose of its biosolids at a facility permitted under Order No. 2000-10-DWQ or at a similar facility permitted under individual WDRs. No matter where the biosolids are taken, the Discharger must comply with all sampling and analytical requirements of the entity that accepts the waste.
7. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR Part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger shall comply with the standards and time schedules contained in 40 CFR Part 503 whether or not they have been incorporated into this Order.

F. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

G. Provisions

1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code:
 - a. By **1 May 2001**, the Discharger shall submit a Groundwater Monitoring Workplan. Sufficient monitoring wells shall be installed to evaluate water quality upgradient and downgradient of the storage reservoirs. The workplan shall be consistent with and include the items listed in the first section of Attachment B.
 - b. By **15 May 2001**, the Discharger shall submit a report, prepared and signed by a California Registered Engineer or Geologist, regarding improvements to the new storage reservoir. The report shall (a) describe the re-grading work to remove the bench, (b) describe the type of liner installed and the installation procedures, QA/QC measures, (c) provide calculations showing that the liner shall ensure compliance with the Groundwater Limitation in this Order, and (d) provide calculations showing the volume of waste which may be stored in the reservoir, assuming a two-foot freeboard level.
 - c. By **1 September 2001**, the Discharger shall submit a Groundwater Monitoring Well Installation Report. The Report shall describe all monitoring devices installed and contain as-built diagrams, and shall include all items listed in the second section of Attachment B.
 - d. By **1 June 2001**, the Discharger shall submit a report, prepared and signed by a registered engineer that certifies construction of the spray disposal field. The report at a minimum must address the total acreage of the constructed sprayfield, the number of lines and sprinkler heads, tailwater collection system, setbacks, location of the sprayfield pump house, and specs for the sprayfield pumps.
 - e. By **15 June 2001**, the Discharger shall submit a Wastewater Disposal Operations Plan, prepared and signed by a registered engineer that will completely describe the capacity and function of each storage reservoir, monitoring devices and locations. The Plan shall also describe the number

of acres used, how, where, and when reclaimed wastewater is disposed to the sprayfield. The Wastewater Disposal Operations Plan shall be accompanied with a detailed set of maps for the wastewater treatment plant and the sprayfield.

- f. By **1 July 2001**, the Discharger shall submit a Solids Management Plan for describing the permanent disposal of biosolids, the long-term management of biosolids, and all other non-effluent wastes generated by the treatment process. The Solids Management Plan shall provide a detailed program and schedule for permanent disposal of all solid wastes that will be generated in the future. Information provided shall include methods and locations of temporary on-site storage (if used), Best Management Practices for on-site handling and storage of solid waste, means of disposal, frequency of disposal, and disposal site (as applicable).
 - g. By **1 September 2001**, the Discharger shall submit an engineering report, prepared and signed by a registered engineer, which certifies the 0.600 million-gallon storage reservoir has been lined and is ready for use. The report shall also define the total storage capacity of the reservoir and account for two feet of freeboard. All surveys and supplemental calculations shall be included in the report.
- 2. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-056, which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
- 3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
- 4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
- 5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.

6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 March 2001.

GARY M. CARLTON, Executive Officer